Projector-Resolved Seed Calculus IX

Arithmetic Packet Calculus: Euler Families, Supercongruences, and Weighted p-Adic Reconstruction

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Abstract

This paper develops the arithmetic realization of the projector-resolved seed calculus. Integer and rational sequences are treated as coefficient objects acted on by residue projectors, shifts, finite differences, exponential-generating-function convolution, triangular coordinate changes, and p-adic completion. The analytic families introduced earlier are reorganized in a single proof-normalized arithmetic notation. Congruential Euler packets are shown to satisfy exact lower-triangular inverse systems with explicit denominator clearing. For every odd prime p, the alternating odd-power coordinates

$$S_n(p) = \sum_{a=0}^{p-1} (-1)^a (2a+1)^n$$

recover the even Euler numbers through an explicit unit-triangular transform whose mth off-diagonal carries p^{2m}. Truncation at depth M therefore yields a uniform congruence modulo p^{2M+2}. The dyadic sector is developed separately. A new exact coordinate system

$$R_n = \frac{3^{n+2} - 1}{4(n+1)(n+2)}$$

for even n is inverted by the coefficients of $z^2/\sinh^2 z$ after the rescaling $z = 2t$. Both coordinate and inverse coefficients are 2-adically integral, and the depth-M reconstruction is uniform modulo 2^{2M+2}. The global odd-prime finite-difference theorem is integrated in the normalization used here: the unnormalized packets $j!C_{mp,j}(n)$ satisfy the all-class bound, the normalized packets satisfy the corresponding factorial-loss bound, and the full bound is recovered without loss in the standard window $0 \le j < p$. This proves the uniform standard-window period and also identifies a concrete counterexample to the earlier unrestricted normalized all-residue formulation. Mahler coordinates convert finite differences into coefficient shifts, and companion-matrix jumps produce exact arithmetic subsequence recurrences. Sixteen code-generated figures, an exact formula registry, expanded exact arithmetic checks, and the accompanying formal proof supplement complete the release. Only the canonical arithmetic obstruction packet and the prime-independent arithmetic spine remain conjectural in this paper.

Scope and claim boundary. The paper proves the exact packet identities, triangular inverse laws, odd-prime Euler coordinate theorem, odd-prime supercongruence ladder, weighted dyadic Euler reconstruction, dyadic tail bound, the global odd-prime finite-difference theorem in both unnormalized and normalized packet coordinates, the uniform standard-window period, Mahler difference identities, and recurrence-packet formulas stated below. The unrestricted normalized all-residue bound is not retained: it is false once $j!$ carries positive p-adic valuation, and the correct factorial-loss estimate is stated explicitly. Classical ingredients such as Lucas' theorem, Mahler expansion, Bernoulli numbers, the Chinese remainder theorem, and companion-matrix recurrence theory are identified as inputs. The complete proof of the global odd-prime theorem is included in the release as a formal supplement.

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1 Arithmetic realization of the seed calculus

The common operator chain becomes

coefficient seed $\longrightarrow$ index address $\longrightarrow$ residue projection
$\longrightarrow$ triangular transfer $\longrightarrow$ valuation filtration $\longrightarrow$ completion $\longrightarrow$ reconstruction.

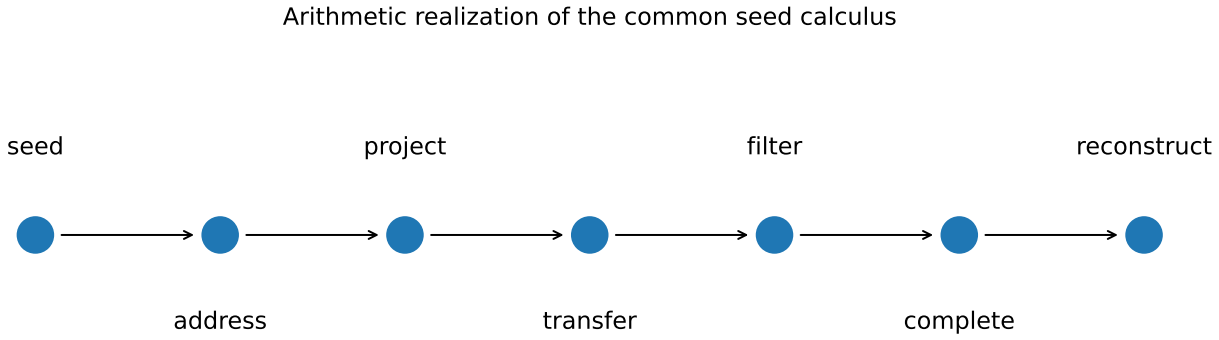


Figure 1: The arithmetic realization. The same coefficient object may be addressed by degree, residue, finite-difference order, recurrence state, or p-adic depth.

Definition 1.1 (Arithmetic seed system)**.** An arithmetic seed system is a tuple

$$\mathcal{S}_{\mathrm{arith}} = (A,\, a_0,\, \mathcal{O},\, \mathcal{R},\, \{P_{N,j}\},\, w,\, \tau_p),$$

where A is a coefficient ring or module, a_0 is a seed coefficient object, $\mathcal{O}$ contains allowed operators such as shift, multiplication, differentiation, finite difference, convolution, and recurrence jump, $\mathcal{R}$ is the relation system, $P_{N,j}$ are residue projectors, w is a valuation or depth weight, and τ_p is a p-adic, product, or formal topology.

For a sequence $a = (a_n)_{n\geq 0}$, define

$$(P_{N,j}a)_n = \mathbf{1}_{n\equiv j \,(\mathrm{mod}\, N)} a_n.$$

For an exponential generating function

$$A(t) = \sum_{n\geq 0} a_n \frac{t^n}{n!},$$

the same projector is

$$(P_{N,j}A)(t) = \sum_{n\equiv j \,(\mathrm{mod}\, N)} a_n \frac{t^n}{n!}.$$

Theorem 1.2 (Exact arithmetic packet reconstruction)**.** *For every $N \geq 1$,*

$$P_{N,i}P_{N,j} = \delta_{ij} P_{N,j}, \qquad \sum_{j=0}^{N-1} P_{N,j} = I.$$

Hence every coefficient sequence and every formal or convergent EGF reconstructs exactly:

$$\boxed{a = \sum_{j=0}^{N-1} P_{N,j} a, \qquad A(t) = \sum_{j=0}^{N-1} (P_{N,j} A)(t).}$$

If $N \mid M$, then

$$\boxed{P_{N,j} = \sum_{\substack{0 \le r < M \\ r \equiv j \pmod N}} P_{M,r}.}$$

Proof. Each nonnegative index lies in exactly one residue class modulo N. The first two identities follow coefficient by coefficient. When $N \mid M$, the residue class j modulo N is the disjoint union of the residues r modulo M satisfying $r \equiv j \pmod N$. □

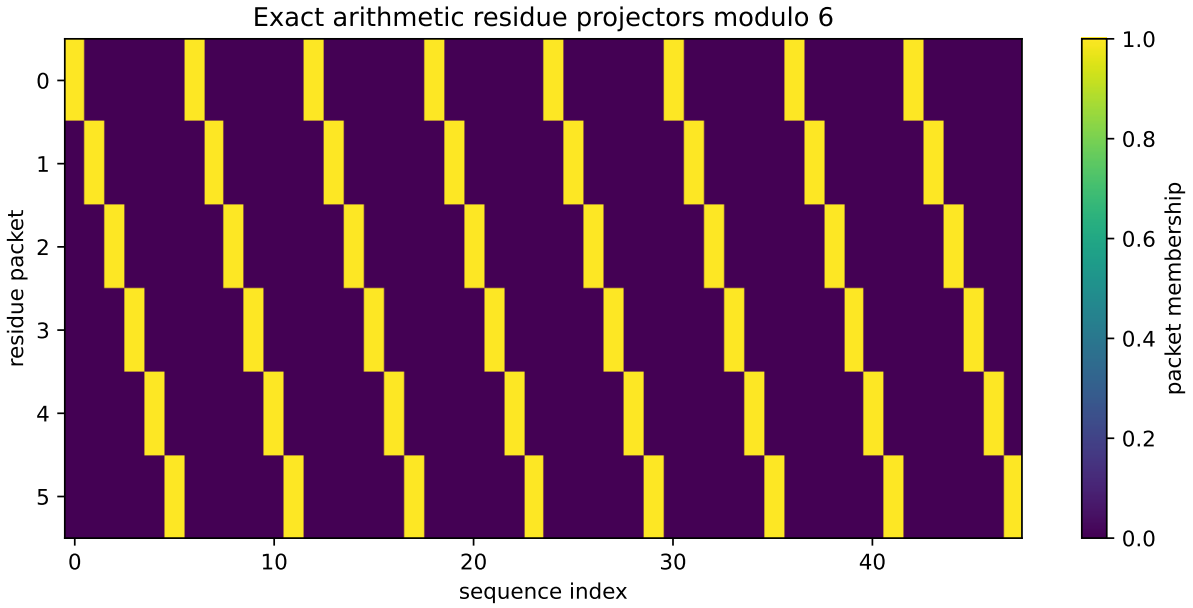


Figure 2: The coefficient selector matrix for six arithmetic residue packets. Every column belongs to exactly one packet.

Remark 1.3 (What is arithmetic here). The projector is algebraic. Arithmetic enters when the packets carry divisibility, congruence, recurrence, or completion information. The same residue split may be tautological over $\mathbb{Q}$ and decisive over $\mathbb{Z}_p$ because different packet transfers acquire different valuations.

2 Unit-triangular arithmetic coordinates

Let

$$A(t) = \sum_{n \ge 0} A_n \frac{t^n}{n!}$$

be a coefficient germ and let

$$H(t) = 1 + \sum_{r \ge 1} h_r \frac{t^r}{r!}$$

be a unit EGF. Define new coordinates by

$$S(t) = H(t)A(t).$$

Theorem 2.1 (Unit EGF coordinate change)**.** *Let*

$$K(t) = H(t)^{-1} = \sum_{r\geq 0} k_r \frac{t^r}{r!}, \qquad k_0 = 1.$$

Then

$$\boxed{S_n = \sum_{r=0}^{n} \binom{n}{r} h_r A_{n-r}, \qquad A_n = \sum_{r=0}^{n} \binom{n}{r} k_r S_{n-r}.}$$

The corresponding lower-triangular operator U_H has unit diagonal and is invertible. Under the coordinate change, every packet projector transports by conjugation:

$$\boxed{\widehat{P}_{N,j} = U_H P_{N,j} U_H^{-1}.}$$

The transported projectors remain idempotent, orthogonal, and reconstructive.

Proof. The two coefficient identities are the EGF product rule applied to $S = HA$ and $A = KS$. Since $H(0) = 1$, the diagonal entries are one. Conjugation preserves all polynomial identities satisfied by the projectors. □

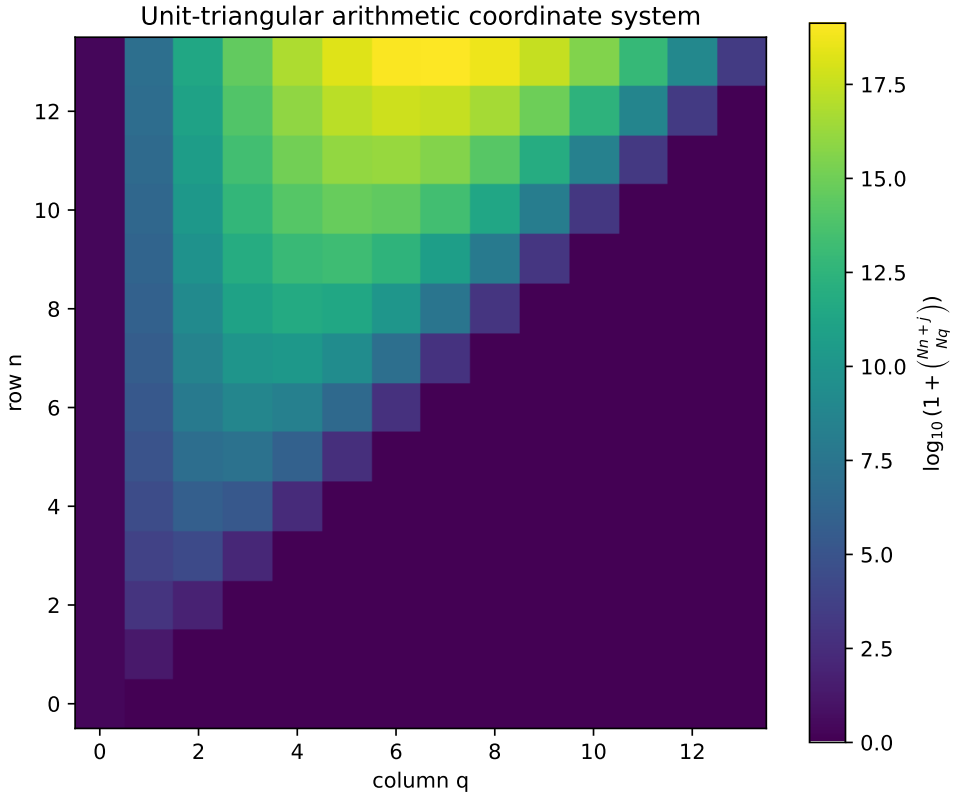


Figure 3: A typical arithmetic coordinate matrix. The triangular geometry makes exact inversion and denominator tracking explicit.

The existence of an invertible matrix alone has little content. The useful coordinate changes are those with an explicit kernel, a finite rule, packet compatibility, and a filtration that controls truncation.

Theorem 2.2 (Filtered triangular reconstruction)**.** *Let p be prime and suppose*

$$A(t) = K_p(t)S(t), \qquad K_p(t) = \sum_{r\geq 0} c_r p^{w(r)} \frac{t^r}{r!},$$

where $c_r \in \mathbb{Z}_p$, $w(r) \in \mathbb{N} \cup \{\infty\}$, and $w(r) \to \infty$. Assume $S_n \in \mathbb{Z}_p$. Then

$$A_n = \sum_{r=0}^{n} \binom{n}{r} c_r p^{w(r)} S_{n-r}.$$

For every precision L,

$$\boxed{A_n \equiv \sum_{\substack{0\leq r\leq n \\ w(r)<L}} \binom{n}{r} c_r p^{w(r)} S_{n-r} \pmod{p^L}.}$$

The modulus depends only on the omitted weight, not on a special row n.

Proof. Every omitted summand contains $p^{w(r)}$ with $w(r) \geq L$. All remaining factors are p-integral, so the omitted tail belongs to $p^L\mathbb{Z}_p$. □

Boundary 2.3 (Coordinate freedom). An arbitrary invertible triangular matrix can encode an arbitrary sequence and produces no meaningful theorem. In this paper a coordinate change is retained only when its kernel is explicit, its diagonal is controlled, its packet action is known, and its filtration produces verifiable divisibility.

3 Congruential Euler packets

Define the exponential section

$$\mathcal{E}_{N,j}(t) = \sum_{n\geq 0} \frac{t^{Nn+j}}{(Nn+j)!}, \qquad 0 \leq j < N.$$

It vanishes to order j. Removing the forced factor gives a unit germ.

Definition 3.1 (Congruential Euler packet)**.** Define

$$\boxed{\mathcal{C}_{N,j}(t) = \frac{t^j/j!}{\mathcal{E}_{N,j}(t)} = \sum_{n\geq 0} C_{N,j}(n) \frac{t^{Nn}}{(Nn)!}.}$$

Theorem 3.2 (Triangular congruential inverse)**.** *The coefficients satisfy $C_{N,j}(0) = 1$ and*

$$\boxed{\sum_{q=0}^{n} \binom{Nn+j}{Nq} C_{N,j}(q) = \delta_{n0}.}$$

For $n \geq 1$,

$$\boxed{C_{N,j}(n) = -\frac{\displaystyle\sum_{q=0}^{n-1} \binom{Nn+j}{Nq} C_{N,j}(q)}{\displaystyle\binom{Nn+j}{Nn}}.}$$

Proof. Multiply $\mathcal{C}_{N,j}(t)$ by $\mathcal{E}_{N,j}(t)$ and compare the coefficient of $t^{Nn+j}/(Nn+j)!$. The diagonal coefficient is nonzero, so the system has a unique solution. □

Proposition 3.3 (Universal denominator clearing)**.** *Set*

$$D_{N,j}(n) = \prod_{r=1}^{n} \binom{Nr+j}{Nr}, \qquad D_{N,j}(0) = 1.$$

Then

$$\boxed{D_{N,j}(n) C_{N,j}(n) \in \mathbb{Z}}$$

for every $n \geq 0$.

Proof. Proceed by induction on n. Every earlier coefficient $C_{N,j}(q)$ has denominator dividing $D_{N,j}(q)$, which divides $D_{N,j}(n-1)$. Multiplying the recurrence by $D_{N,j}(n) = D_{N,j}(n-1)\binom{Nn+j}{Nn}$ clears the new diagonal denominator and all earlier ones. □

Example 3.4 (Euler boundary)**.** For $(N, j) = (2, 0)$,

$$\mathcal{E}_{2,0}(t) = \cosh t, \qquad \mathcal{C}_{2,0}(t) = \operatorname{sech} t.$$

Thus

$$C_{2,0}(n) = E_{2n},$$

where

$$\operatorname{sech} t = \sum_{n \geq 0} E_n \frac{t^n}{n!}.$$

The first even coefficients are

$$1, -1, 5, -61, 1385, -50521, \ldots.$$

Example 3.5 (Bernoulli boundary)**.** For $(N, j) = (2, 1)$,

$$\mathcal{C}_{2,1}(t) = \frac{t}{\sinh t}.$$

If B_n are the Bernoulli numbers with $B_1 = -1/2$, then

$$\boxed{C_{2,1}(n) = 2\left(1 - 2^{2n-1}\right) B_{2n}, \qquad n \geq 0,}$$

where 2^{-1} is interpreted rationally at $n = 0$.

Proof. Use

$$\frac{t}{\sinh t} = \frac{2te^{-t}}{1 - e^{-2t}}$$

and the Bernoulli-polynomial generating function, or compare the standard Bernoulli expansion term by term. □

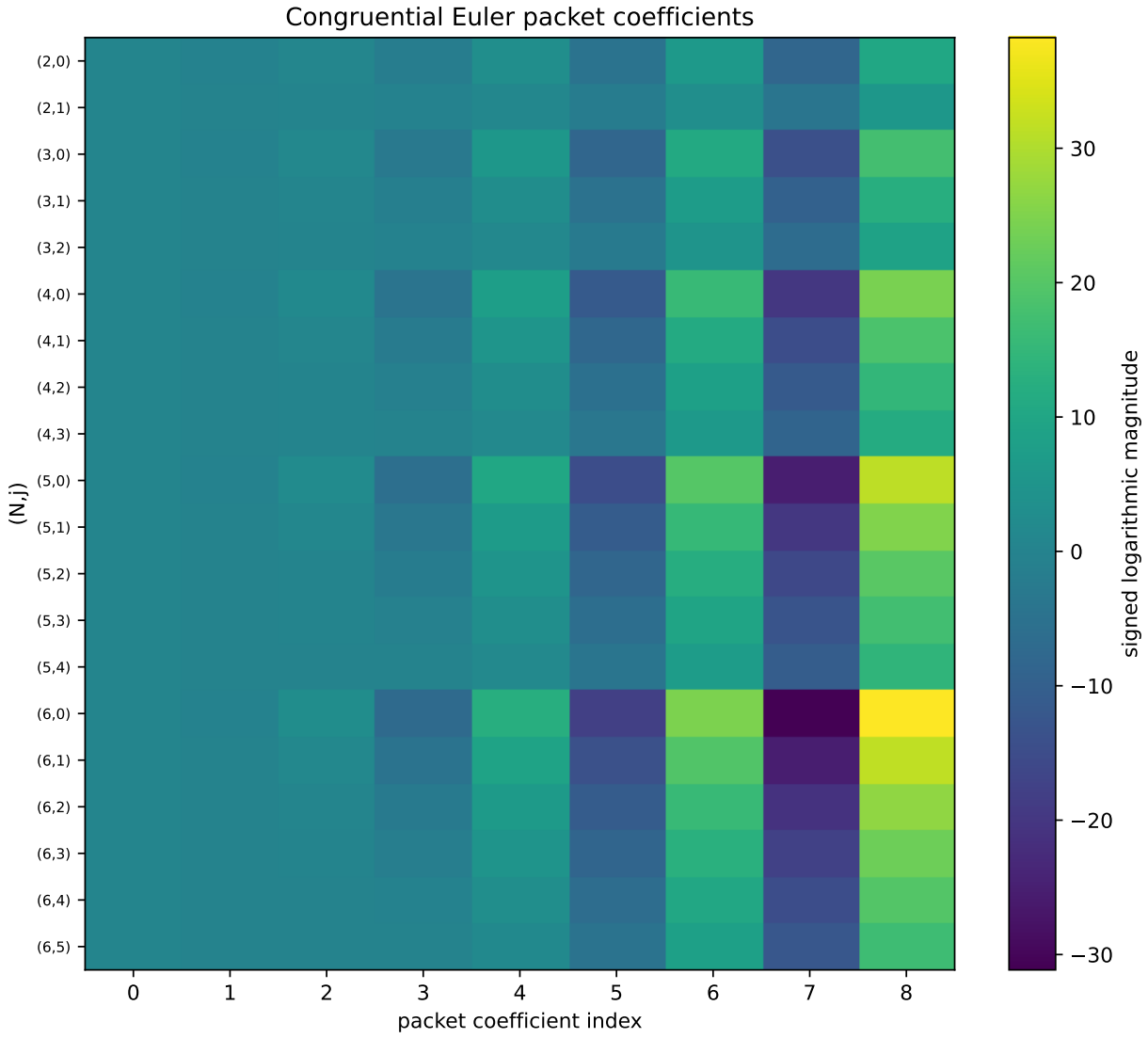


Figure 4: Congruential Euler packets for $2 \leq N \leq 6$. The family contains the Euler and Bernoulli boundaries but is not limited to them.

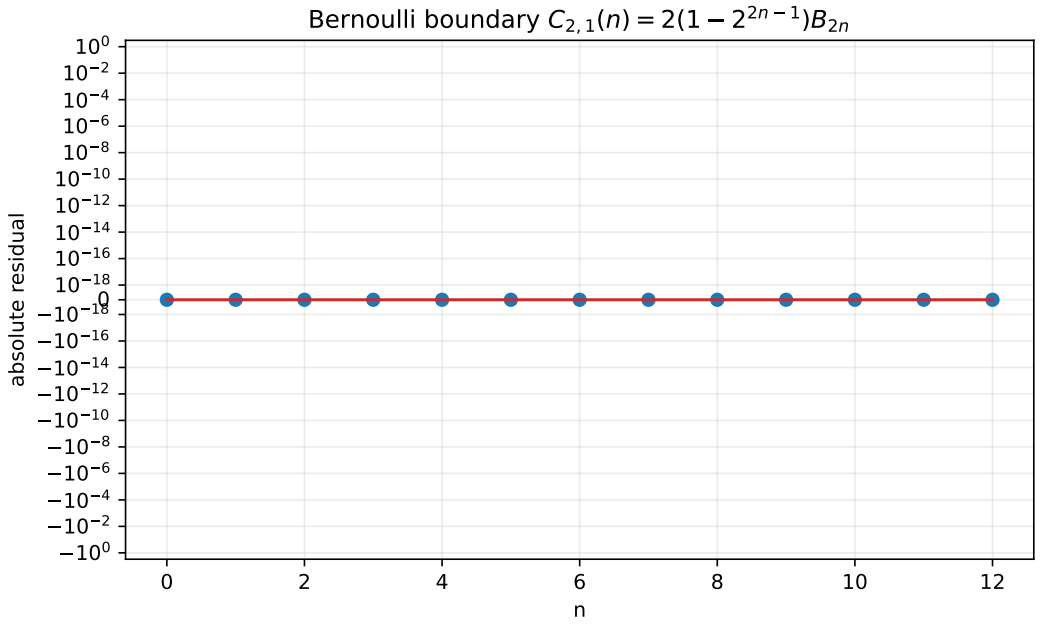


Figure 5: Exact agreement between the triangular packet inverse and the Bernoulli closed form. The plotted residual is identically zero.

4 The uniform odd-prime Euler coordinate theorem

Let p be an odd prime and define the alternating odd-power coordinates

$$\boxed{S_n(p) = \sum_{a=0}^{p-1} (-1)^a (2a+1)^n.}$$

Their EGF is elementary but carries the entire coordinate change.

Lemma 4.1 (Odd-power generating function)**.** *For odd p,*

$$\boxed{\sum_{n\geq 0} S_n(p) \frac{t^n}{n!} = \frac{1+e^{2pt}}{2} \operatorname{sech} t.}$$

Its even part is

$$\boxed{\sum_{n\geq 0} S_{2n}(p) \frac{t^{2n}}{(2n)!} = \cosh^2(pt) \operatorname{sech} t.}$$

Proof. Sum the finite geometric progression

$$\sum_{a=0}^{p-1} (-1)^a e^{(2a+1)t} = e^t \frac{1-(-e^{2t})^p}{1+e^{2t}}.$$

Because p is odd, the numerator is $1+e^{2pt}$. The first formula follows from $2e^t/(1+e^{2t}) = \operatorname{sech} t$. Averaging the expression at t and $-t$ gives the even part, and

$$\frac{1+\cosh(2pt)}{2} = \cosh^2(pt).$$

□

Let Z_n denote the Euler zigzag numbers,

$$\sec x + \tan x = \sum_{n\geq 0} Z_n \frac{x^n}{n!}.$$

Differentiating $\tanh x$ gives

$$\boxed{\operatorname{sech}^2 x = \sum_{m\geq 0} (-1)^m Z_{2m+1} \frac{x^{2m}}{(2m)!}.}$$

Theorem 4.2 (Odd-prime Euler reconstruction)**.** *For every odd prime p and every even $n \geq 0$,*

$$\boxed{E_n = \sum_{m=0}^{n/2} (-1)^m Z_{2m+1} \binom{n}{2m} p^{2m} S_{n-2m}(p).}$$

The coordinate matrix on the even indices is lower triangular with diagonal one.

Proof. The even-part identity gives

$$\operatorname{sech} t = \operatorname{sech}^2(pt) \sum_{r\geq 0} S_{2r}(p) \frac{t^{2r}}{(2r)!}.$$

Insert the EGF expansion of $\operatorname{sech}^2(pt)$ and compare the coefficient of $t^n/n!$. □

Corollary 4.3 (Uniform odd-prime supercongruence ladder)**.** *For every $M \geq 0$,*

$$\boxed{E_n \equiv \sum_{m=0}^{\min(M,n/2)} (-1)^m Z_{2m+1} \binom{n}{2m} p^{2m} S_{n-2m}(p) \pmod{p^{2M+2}}.}$$

The statement is uniform in the even row n.

Proof. Every omitted term has $m \geq M+1$, hence contains p^{2m} with $2m \geq 2M+2$. All remaining factors are integers. □

Example 4.4 (The row $n = 8$ at $p = 5$)**.** The necessary coordinates are

$$S_0 = 1, \quad S_2 = 49, \quad S_4 = 4705, \quad S_6 = 428689, \quad S_8 = 37665985.$$

The exact row is

$$\begin{aligned} E_8 = S_8 - 56p^2 S_6 + 1120p^4 S_4 \\ - 7616p^6 S_2 + 7936p^8 S_0. \end{aligned}$$

At $p = 5$ this equals 1385. Truncating after $M = 2$ gives

$$T_2 = 2731001385,$$

and

$$E_8 - T_2 = -2731000000, \qquad v_5(E_8 - T_2) = 6.$$

Thus the guaranteed modulus 5^6 is attained exactly in this row.

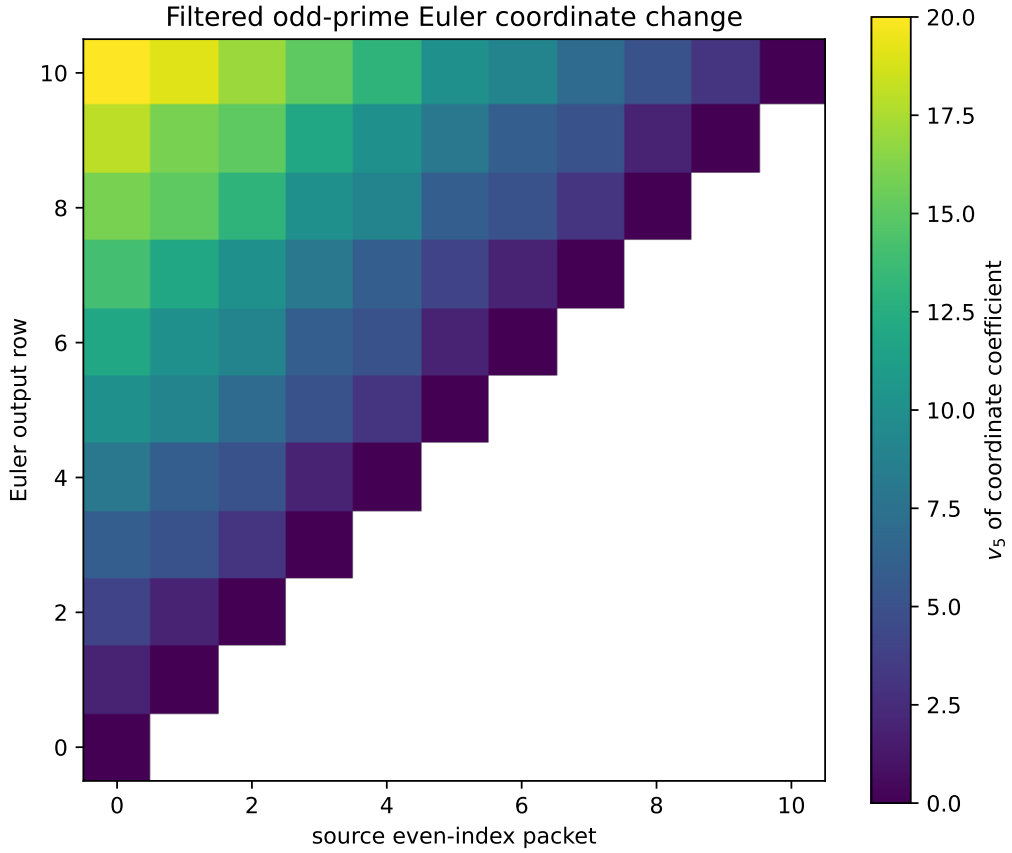


Figure 6: The 5-adic valuation of the odd-power-to-Euler coordinate matrix. Moving one packet step away from the diagonal contributes at least two powers of 5.

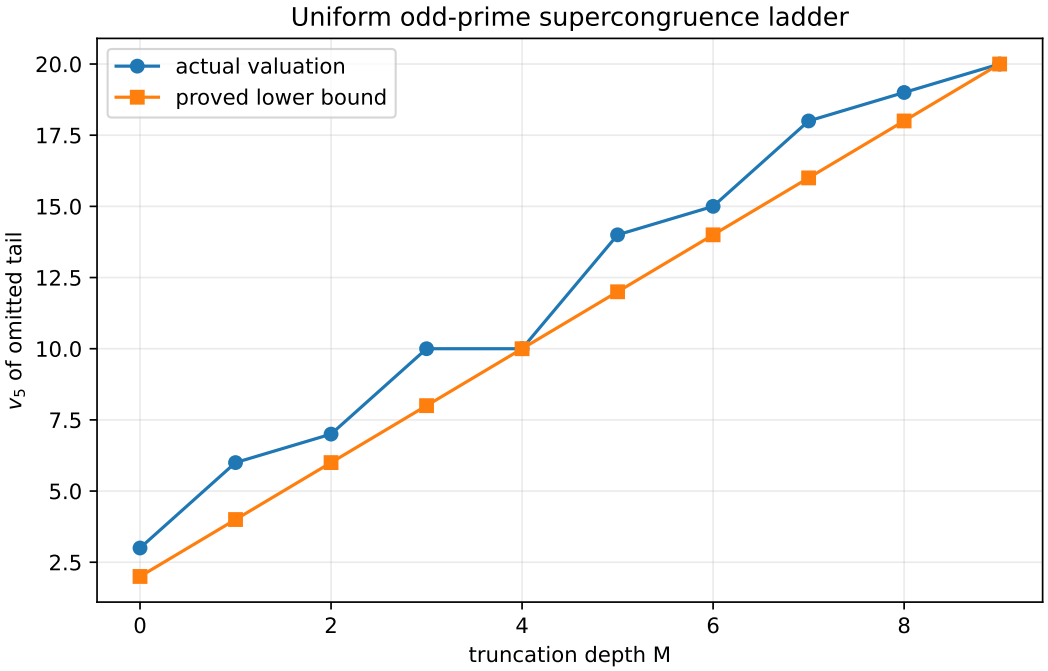


Figure 7: Actual omitted-tail valuations and the proved lower bound for one odd-prime Euler row. The bound depends on truncation depth, not on a specially selected row.

5 Weighted dyadic Euler reconstruction

The odd-prime formula cannot be specialized by writing $p = 2$. The dyadic unit group, sign packet, denominators, and natural filtration are different. We therefore construct a separate coordinate system.

For even $n \geq 0$, define

$$\boxed{R_n = \frac{3^{n+2} - 1}{4(n+1)(n+2)}.}$$

Its EGF is

$$\boxed{\sum_{\substack{n \geq 0 \\ n \text{ even}}} R_n \frac{t^n}{n!} = \frac{\sinh(2t)\sinh t}{2t^2}.}$$

Indeed,

$$\frac{\sinh(2t)\sinh t}{2t^2} = \frac{\cosh(3t) - \cosh t}{4t^2}.$$

Define β_m by the ordinary-power expansion

$$\boxed{\frac{z^2}{\sinh^2 z} = \sum_{m \geq 0} \beta_m z^{2m}.}$$

Equivalently,

$$\beta_0 = 1, \qquad \boxed{\beta_m = -\frac{2^{2m}(2m-1)B_{2m}}{(2m)!} \quad (m \geq 1).}$$

The first values are

$$1, -\frac{1}{3}, \frac{1}{15}, -\frac{2}{189}, \frac{1}{675}, \ldots.$$

Lemma 5.1 (Dyadic integrality of the inverse coefficients)**.** *For every $m \geq 0$,*

$$\boxed{\beta_m \in \mathbb{Z}_2.}$$

Proof. Write

$$\left(\frac{\sinh z}{z}\right)^2 = \frac{\cosh(2z) - 1}{2z^2} = \sum_{m \geq 0} \frac{2^{2m+1}}{(2m+2)!} z^{2m}.$$

For every m,

$$v_2((2m+2)!) \leq 2m + 1,$$

so every coefficient belongs to $\mathbb{Z}_2$. The constant coefficient is one. The inverse of a unit power series over $\mathbb{Z}_2$ again has coefficients in $\mathbb{Z}_2$, and that inverse is $z^2/\sinh^2 z$. □

Lemma 5.2 (Dyadic integrality of the source)**.** *For every even $n \geq 0$,*

$$\boxed{R_n \in \mathbb{Z}_2.}$$

Proof. Let $k = n + 2$, which is even. The lifting-the-exponent identity gives

$$v_2(3^k - 1) = v_2(3 - 1) + v_2(3 + 1) + v_2(k) - 1 = 2 + v_2(k).$$

The denominator $4(n + 1)(n + 2)$ has the same 2-adic valuation because $n + 1$ is odd. Hence the quotient is 2-adically integral. □

Theorem 5.3 (Exact weighted dyadic Euler coordinate theorem)**.** *For every even $n \geq 0$,*

$$\boxed{E_n = \sum_{m=0}^{n/2} \beta_m 2^{2m} (n)_{2m} R_{n-2m},}$$

where

$$(n)_{2m} = \frac{n!}{(n - 2m)!}.$$

Proof. Substitute $z = 2t$ into the inverse series:

$$\frac{4t^2}{\sinh^2(2t)} = \sum_{m \geq 0} \beta_m 2^{2m} t^{2m}.$$

Multiplying by the source EGF gives

$$\frac{4t^2}{\sinh^2(2t)} \frac{\sinh(2t) \sinh t}{2t^2} = \frac{2 \sinh t}{\sinh(2t)} = \operatorname{sech} t.$$

Coefficient extraction converts the ordinary factor t^{2m} into the falling factorial $(n)_{2m}$. □

Corollary 5.4 (Weighted dyadic supercongruence)**.** *For every $M \geq 0$,*

$$\boxed{E_n \equiv \sum_{m=0}^{\min(M, n/2)} \beta_m 2^{2m} (n)_{2m} R_{n-2m} \pmod{2^{2M+2}}}$$

in $\mathbb{Z}_2$.

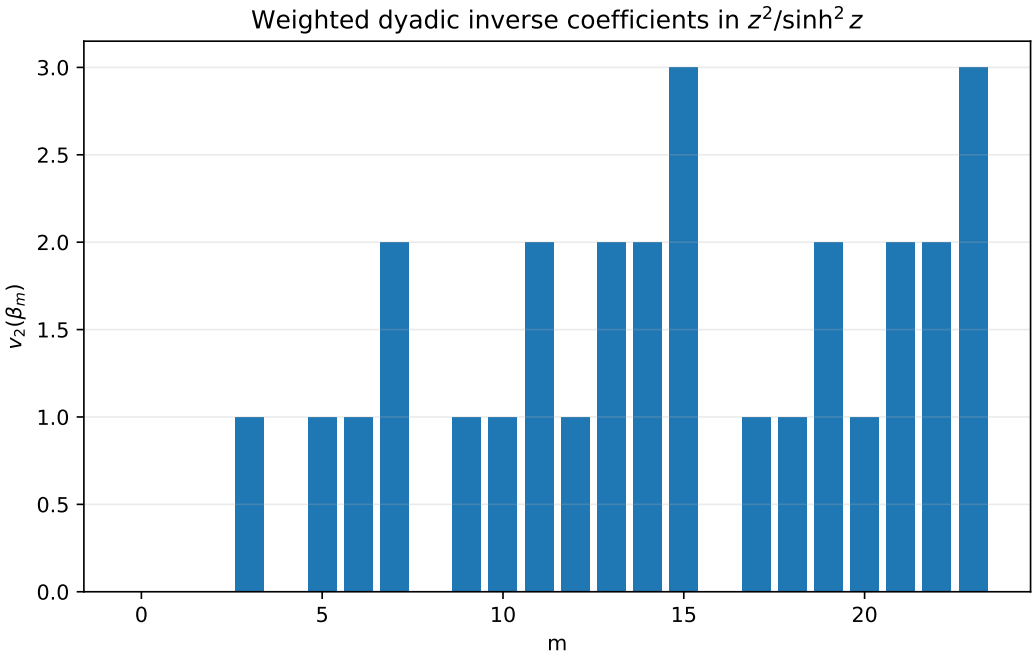


Figure 8: The inverse coefficients are 2-adically integral. Their extra valuations strengthen some rows, but the uniform theorem uses only the visible factor 2^{2m}.

Proof. The preceding lemmas show that β_m, $(n)_{2m}$, and R_{n-2m} are 2-adically integral. Each omitted term has $m \geq M+1$ and therefore carries the factor $2^{2m} \in 2^{2M+2}\mathbb{Z}_2$. □

Example 5.5 (The dyadic row $n = 8$)**.** The exact decomposition is

$$\begin{aligned} E_8 &= \frac{7381}{45} - \frac{6560}{3} + \frac{163072}{15} - \frac{204800}{9} + \frac{229376}{15} \\ &= 1385. \end{aligned}$$

The first truncation has error

$$1385 - \frac{7381}{45} = \frac{54944}{45}, \qquad v_2\left(\frac{54944}{45}\right) = 5,$$

which is stronger than the uniform depth-zero bound 2. After three transfer levels, the error is

$$\frac{229376}{15}, \qquad v_2\left(\frac{229376}{15}\right) = 15,$$

against the guaranteed bound 8.

6 Why the dyadic and odd-adic coordinates differ

Classical Input 6.1 (Unit-group decomposition)*.* For odd p,

$$\mathbb{Z}_p^\times \cong \mu_{p-1} \times (1 + p\mathbb{Z}_p).$$

At $p = 2$,

$$\mathbb{Z}_2^\times \cong \{\pm 1\} \times (1 + 4\mathbb{Z}_2).$$

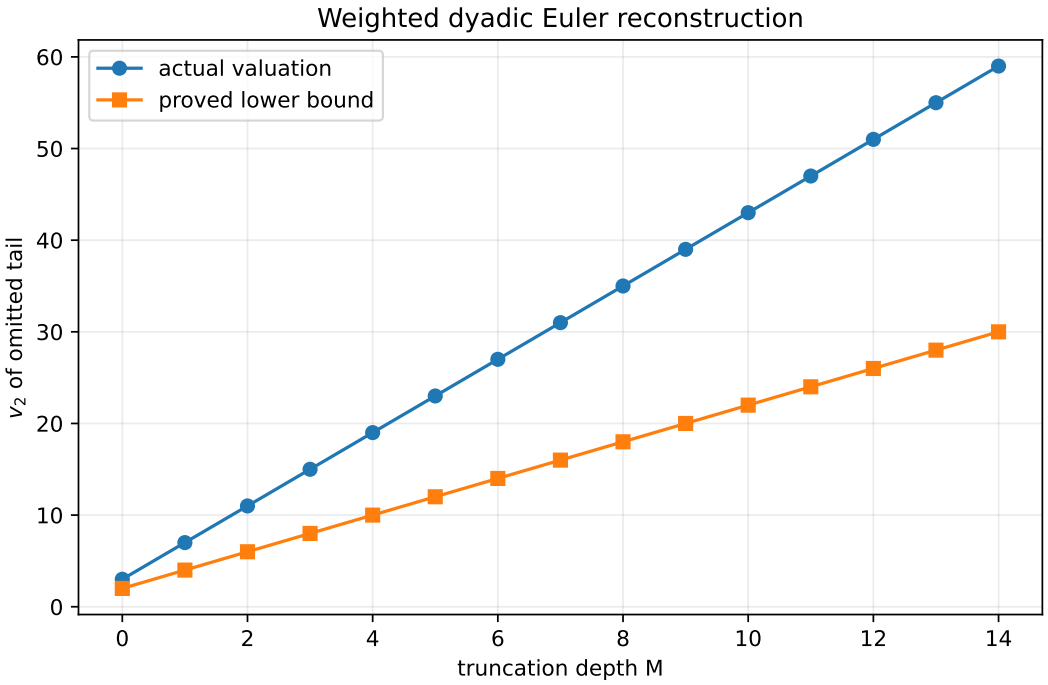


Figure 9: The weighted dyadic ladder. The proved lower bound is uniform; additional valuation comes from the inverse coefficients and falling factorials.

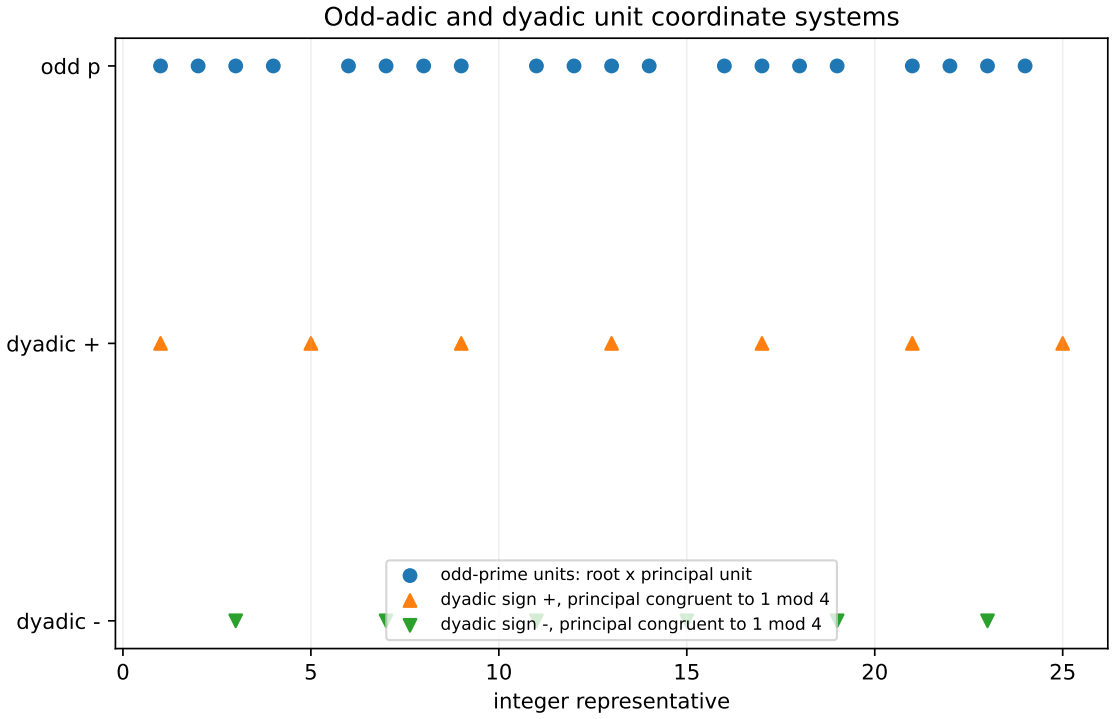


Figure 10: Finite representatives of the odd-prime and dyadic unit decompositions. The first dyadic decision is a sign packet; the principal unit begins modulo 4.

The analytic principal-unit coordinate therefore begins one level deeper in the dyadic case, and the sign packet is invisible modulo 2.

Proposition 6.2 (Finite unit decomposition certificates)**.** *Let p be odd and $L \geq 1$. Every unit $a \pmod{p^L}$ admits a factorization*

$$a \equiv \omega(a)\langle a\rangle \pmod{p^L},$$

where

$$\omega(a)^{p-1} \equiv 1 \pmod{p^L}, \qquad \langle a\rangle \equiv 1 \pmod{p}.$$

For $L \geq 2$, every odd $a \pmod{2^L}$ admits

$$a \equiv \varepsilon(a)\langle a\rangle_2 \pmod{2^L},$$

where

$$\varepsilon(a) \in \{\pm 1\}, \qquad \langle a\rangle_2 \equiv 1 \pmod{4}.$$

Proof. For odd p, reduce a modulo p and consider $X^{p-1} - 1$. Every nonzero residue is a simple root because $(p-1)X^{p-2}$ is a unit modulo p. Hensel lifting therefore gives a unique root $\omega(a)$ modulo p^L congruent to a modulo p. Setting $\langle a\rangle = a\omega(a)^{-1}$ gives the required principal unit.

For $p = 2$, put $\varepsilon(a) = 1$ when $a \equiv 1 \pmod 4$ and $\varepsilon(a) = -1$ when $a \equiv 3 \pmod 4$. Then $\langle a\rangle_2 = \varepsilon(a)a$ is congruent to 1 modulo 4, and $a = \varepsilon(a)\langle a\rangle_2$ because $\varepsilon(a)^2 = 1$. The sign is uniquely determined by the residue modulo 4. □

Remark 6.3. The dyadic distinction is not cosmetic. Character averages, sign projectors, logarithmic coordinates, trace splittings, and spectral resultants can all introduce powers of two that are absent at odd primes. The weighted theorem in the previous section avoids pretending that one prime-independent formula controls every integral lattice.

7 Mahler coordinates and finite-difference packets

Let $\Delta f(n) = f(n+1) - f(n)$. The binomial polynomials are adapted to this operator:

$$\Delta\binom{x}{r} = \binom{x}{r-1}.$$

Classical Input 7.1 (Mahler coordinate system). A suitable continuous function on $\mathbb{Z}_p$ has an expansion

$$f(x) = \sum_{r\geq 0} c_r \binom{x}{r},$$

with $c_r \to 0$ p-adically. At the finite polynomial level used below, the identity is purely algebraic and requires no convergence theorem.

Theorem 7.2 (Finite Mahler reconstruction)**.** *For values $f(0), \ldots, f(N)$, define*

$$c_r = \Delta^r f(0), \qquad 0 \leq r \leq N.$$

Then for every $0 \leq n \leq N$,

$$\boxed{f(n) = \sum_{r=0}^{n} c_r \binom{n}{r}.}$$

More generally,

$$\boxed{\Delta^s f(n) = \sum_{r\geq s} c_r \binom{n}{r-s}.}$$

Proof. The Newton interpolation formula gives the first identity. Applying Δ^s and using the binomial shift law gives the second. □

Corollary 7.3 (Weighted difference bound)**.** *If $c_r \in \mathbb{Z}_p$ and $v_p(c_r) \geq w(r)$, then*

$$\boxed{v_p(\Delta^s f(n)) \geq \min_{\substack{r \geq s \\ \binom{n}{r-s} \neq 0}} \left(w(r) + v_p\binom{n}{r-s} \right).}$$

In particular,

$$v_p(\Delta^s f(n)) \geq \inf_{r \geq s} w(r).$$

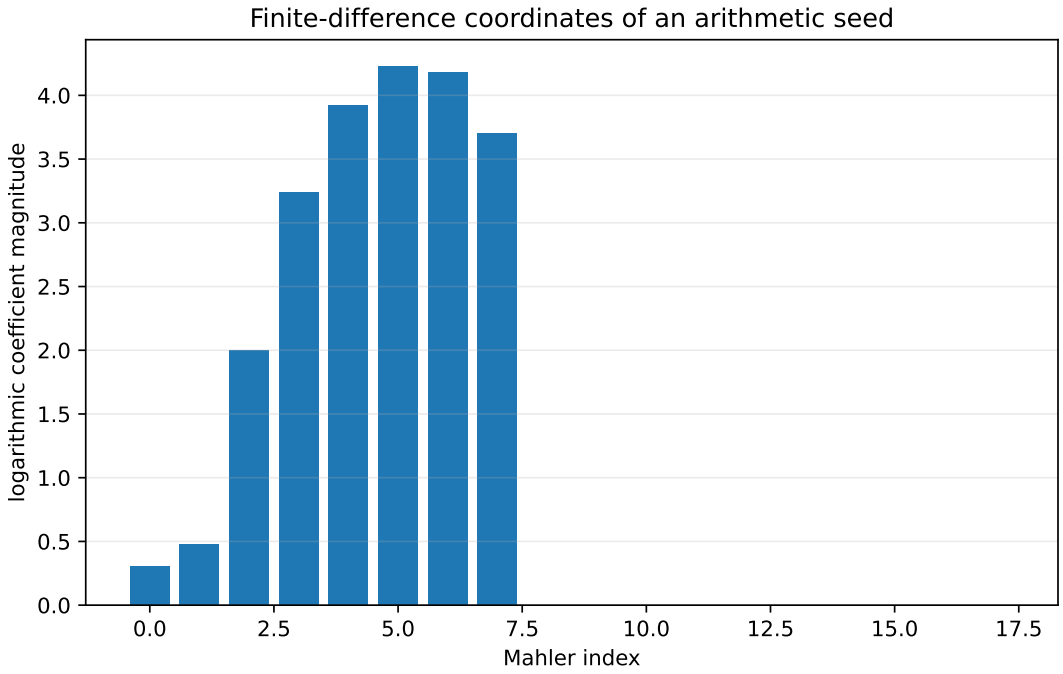


Figure 11: Finite-difference coordinates of a polynomial arithmetic seed. The coefficient tail vanishes exactly after the polynomial degree.

Theorem 7.4 (Unit-mode valuation gain)**.** *Let p be prime, let $u \in \mathbb{Z}_p^\times$, and set $h = p - 1$ for odd p. For*

$$f(n) = u^n,$$

we have

$$\boxed{\Delta_h^s f(n) = u^n (u^{p-1} - 1)^s.}$$

Hence

$$\boxed{v_p(\Delta_h^s f(n)) \geq s.}$$

The same lower bound holds for every finite $\mathbb{Z}_p$-linear combination of such unit modes.

Proof. One h-step difference multiplies u^n by $u^h - 1$. Iteration gives the exact formula. Fermat reduction implies $u^{p-1} - 1 \in p\mathbb{Z}_p$. □

Theorem 7.5 (Global odd-prime finite-difference theorem in packet normalization)**.** *Let p be odd, let $m \mid p-1$, put*

$$N = mp, \qquad h = \frac{p-1}{m}, \qquad E_{Nn}^{(N,j)} = j!C_{N,j}(n),$$

and let $0 \le j < N$. For every $n, k \ge 0$ and every $u \ge 1$,

$$\boxed{E_{Nn}^{(N,j)} \in \mathbb{Z}_p, \qquad v_p\Big(\Delta_{uh}^k E_{Nn}^{(N,j)}\Big) \ge \min\{k, mn\}.}$$

Consequently,

$$\boxed{v_p\Big(\Delta_{uh}^k C_{N,j}(n)\Big) \ge \min\{k, mn\} - v_p(j!).}$$

In the standard window $0 \le j < p$, $j!$ is a p-adic unit and the normalized packet therefore satisfies the full bound

$$v_p\Big(\Delta_{uh}^k C_{N,j}(n)\Big) \ge \min\{k, mn\}.$$

Proof. The complete Dwork–Hurwitz–Mahler proof for the unnormalized coefficients $E_{Nn}^{(N,j)}$ is supplied in `supplements/Odd_Prime_Congruential_Euler_Finite_Difference_Proof.tex` and its compiled PDF. The normalization identity $E_{Nn}^{(N,j)} = j!C_{N,j}(n)$ follows directly by comparing

$$\frac{t^j}{H_{N,j}(t)} \quad \text{with} \quad \frac{t^j/j!}{H_{N,j}(t)}.$$

Finite differences commute with multiplication by the constant $j!$, so subtracting $v_p(j!)$ from the unnormalized valuation gives the normalized estimate. For $j < p$, $v_p(j!) = 0$. □

8 Standard windows and Lucas reduction

Let p be prime, $m \ge 1$, $0 \le j < p$, and set $N = mp$. The congruential Euler recurrence becomes

$$\sum_{q=0}^{n} \binom{mpn + j}{mpq} C_{mp,j}(q) = \delta_{n0}.$$

Classical Input 8.1 (Lucas theorem). If a_i, b_i are the base-p digits of a, b, then

$$\binom{a}{b} \equiv \prod_i \binom{a_i}{b_i} \pmod{p}.$$

Theorem 8.2 (Standard-window row reduction)**.** *For every $0 \le q \le n$,*

$$\boxed{\binom{mpn + j}{mpq} \equiv \binom{mn}{mq} \pmod{p}.}$$

In particular,

$$\boxed{\binom{mpn + j}{mpn} \equiv 1 \pmod{p},}$$

so the standard-window diagonal is a p-adic unit.

Proof. Write

$$mpn + j = p(mn) + j, \qquad mpq = p(mq) + 0.$$

Lucas' theorem separates the low digit:

$$\binom{mpn + j}{mpq} \equiv \binom{j}{0}\binom{mn}{mq} \equiv \binom{mn}{mq} \pmod{p}.$$

The diagonal follows by taking $q = n$. □

Corollary 8.3 (Residue independence in the standard window)**.** *Let $e_n^{(m,p)}$ be the unique sequence modulo p satisfying*

$$\sum_{q=0}^{n} \binom{mn}{mq} e_q^{(m,p)} = \delta_{n0}.$$

Then

$$\boxed{C_{mp,j}(n) \equiv e_n^{(m,p)} \pmod{p}}$$

for every $0 \le j < p$. Thus the first residual inverse is independent of the window address j.

Proof. The row-reduction theorem turns every standard-window recurrence into the same lower-triangular system modulo p. Its diagonal is one, so the solution is unique. □

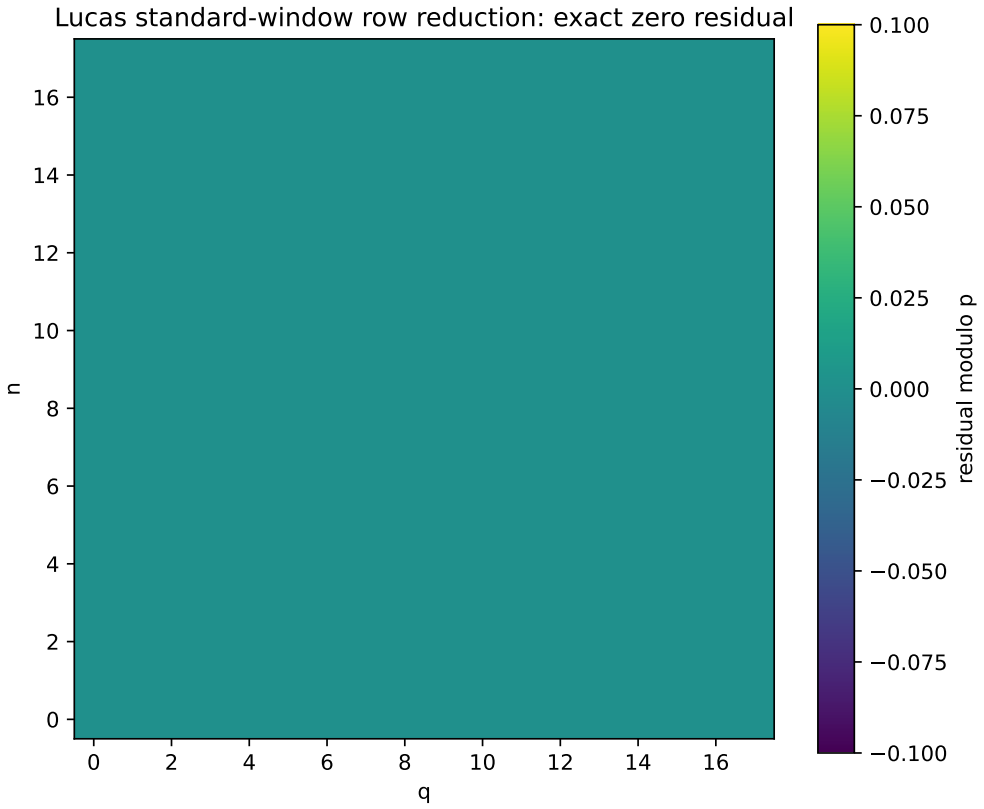


Figure 12: The residual difference between the full standard-window matrix and its Lucas-reduced matrix is identically zero.

Let $m \mid p - 1$ and $h = (p - 1)/m$. The reduced inverse exhibits an apparent tail period h in every tested case.

Numerical Certificate 8.4 (Finite standard-window period certificate). For every divisor $m \mid p-1$, every prime

$$p \in \{3, 5, 7, 11, 13, 17, 19\},$$

and all computed rows through $n = 100$, exact modular extraction gives

$$e_{n+h}^{(m,p)} = e_n^{(m,p)} \qquad (n \geq 1).$$

Thirty parameter pairs were tested. This is a finite certificate, not the proof of a uniform theorem for all primes and all rows.

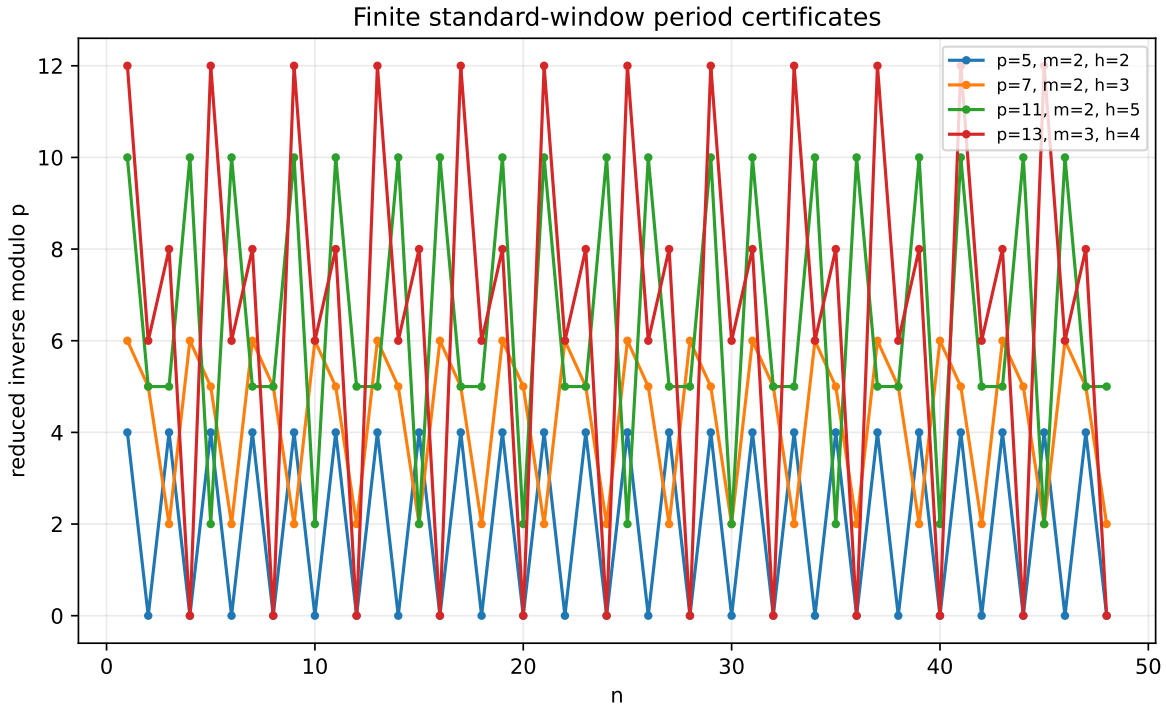


Figure 13: Several exact finite-field period certificates. Their use as evidence is separated from the proved Lucas reduction.

Corollary 8.5 (Uniform standard-window period)**.** *If p is odd, $m \mid p-1$, and $h = (p-1)/m$, then the reduced inverse satisfies*

$$\boxed{e_{n+h}^{(m,p)} = e_n^{(m,p)} \quad \text{in } \mathbb{F}_p, \qquad n \geq 1.}$$

Proof. Choose any $0 \leq j < p$. The global finite-difference theorem with $k = u = 1$ gives

$$C_{mp,j}(n+h) - C_{mp,j}(n) \in p\mathbb{Z}_p \qquad (n \geq 1).$$

The residue-independence corollary identifies the reduction of every such normalized packet with $e_n^{(m,p)}$, proving the stated period. □

9 Arithmetic recurrence packets

A linear recurrence is naturally a first-order operator system. Let

$$v_{n+1} = Cv_n$$

with $C \in M_d(R)$. For a step N and residue j, the packet state is

$$u_m^{(j)} = v_{Nm+j}.$$

Theorem 9.1 (Companion jump theorem)**.** *The packet states satisfy*

$$\boxed{u_m^{(j)} = C^j (C^N)^m v_0.}$$

Consequently every packet subsequence satisfies the order-d recurrence determined by the characteristic polynomial of C^N.

Proof. The first identity is $v_n = C^n v_0$ with $n = Nm + j$. The Cayley-Hamilton theorem applied to C^N produces the packet recurrence. □

Corollary 9.2 (Fibonacci packet recurrence)**.** *Let F_n and L_n denote the Fibonacci and Lucas numbers. For*

$$u_m = F_{Nm+j},$$

we have

$$\boxed{u_{m+2} = L_N u_{m+1} - (-1)^N u_m.}$$

Proof. The Fibonacci companion matrix has eigenvalues α, β with $\alpha\beta = -1$. The jump matrix C^N has trace L_N and determinant $(-1)^N$. □

Example 9.3. For $N = 3$,

$$L_3 = 4, \qquad (-1)^3 = -1,$$

so every residue packet satisfies

$$\boxed{u_{m+2} = 4u_{m+1} + u_m.}$$

For $j = 2$, the packet begins

$$1, 5, 21, 89, 377, 1597, \ldots.$$

10 Dyadic-odd arithmetic polyphase reconstruction

Let

$$N = 2^a M, \qquad M \text{ odd}.$$

The coefficient address modulo N has one dyadic coordinate and one odd coordinate.

Theorem 10.1 (Arithmetic CRT packet factorization)**.** *The map*

$$\boxed{\mathbb{Z}/N\mathbb{Z} \longrightarrow \mathbb{Z}/2^a\mathbb{Z} \times \mathbb{Z}/M\mathbb{Z}, \qquad r \longmapsto (r \bmod 2^a, r \bmod M)}$$

is a bijection. Therefore

$$\boxed{P_{N,r} = P_{2^a,r_2} P_{M,r_o},}$$

where (r_2, r_o) is the CRT address of r.

Proof. This is the Chinese remainder theorem applied to the coprime factors 2^a and M. The projector identity follows because an index satisfies the combined congruence exactly when it satisfies both component congruences. □

Remark 10.2 (Factorization does not imply equal weights). The coefficient address factors exactly, but the integral separation weight need not factor uniformly. A character projector may be integral in the odd coordinate and require additional dyadic depth. The CRT theorem organizes addresses; the weighted theory determines when those addresses are visible on a chosen lattice.

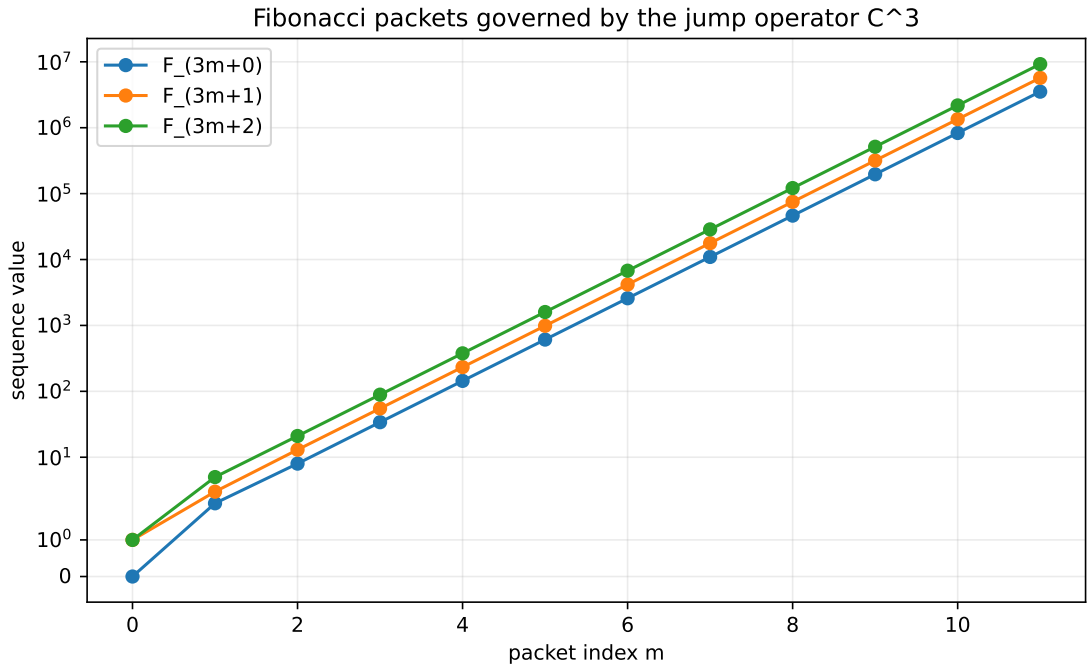


Figure 14: Three Fibonacci packets. The numerical values differ, but all are generated by the same jump operator C^3.

11 Numerical certificates and exact test program

All computations in this paper use exact integers or rational numbers except for the plotting layer. The test program does not use floating-point approximations to decide a congruence.

Table 1: Exact certificate summary.

Certificate	Count or residual	Status
Residue-packet reconstruction	0	exact
Odd-prime reconstruction and truncation checks	180	all passed
Weighted dyadic reconstruction and truncation checks	111	all passed
Standard-window parameter pairs	30	all finite checks passed
Mahler reconstruction residual	0	exact
CRT address checks	192	all passed
Fibonacci packet recurrence residual	0	exact
Automated tests	29/29	passed

Numerical Certificate 11.1 (Odd-prime range). The exact program checked every even $n \le 20$, every depth through $\min(4, n/2)$, and every prime $p \in \{3, 5, 7, 11\}$ in the packaged experiment table. A larger test suite extends the identity and congruence checks through even $n < 34$ and primes through 19.

Numerical Certificate 11.2 (Dyadic range). The exact program checked every even $n \le 40$ and every depth through $\min(5, n/2)$ in the packaged experiment table. The test suite extends exact reconstruction through even $n < 60$ and the uniform tail bound through even $n < 52$.

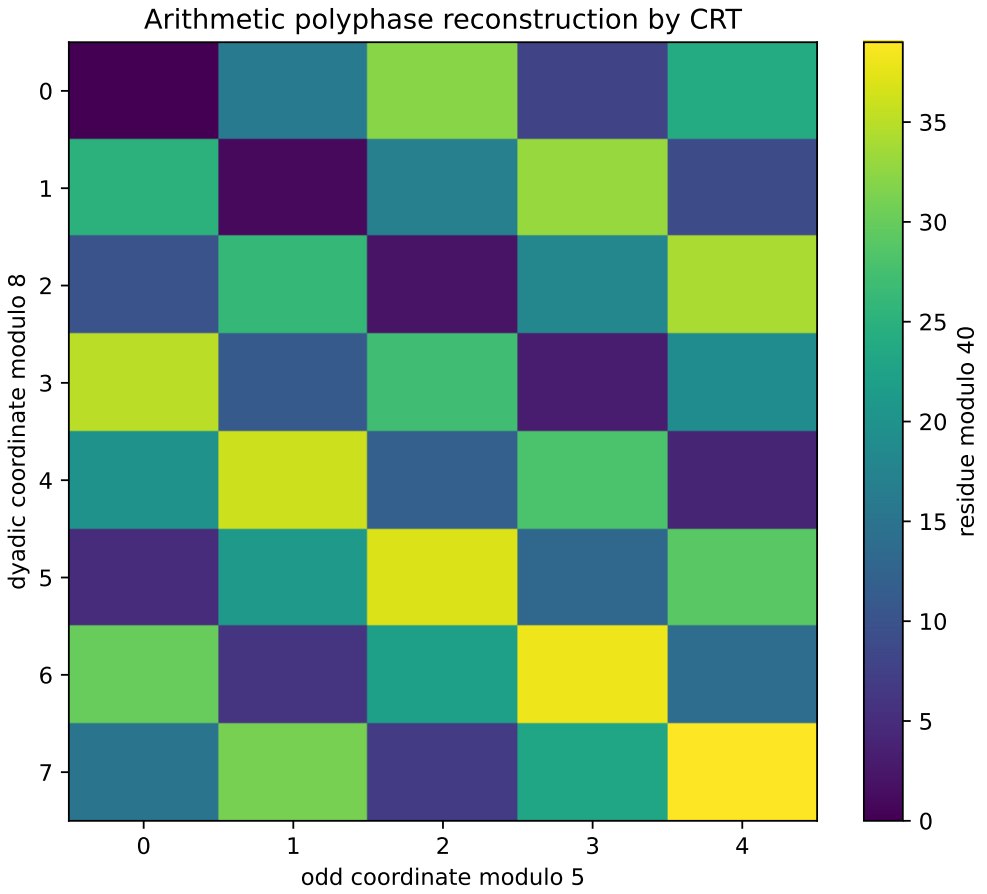


Figure 15: The forty coefficient packets modulo 40 reconstructed from their mod-8 and mod-5 addresses.

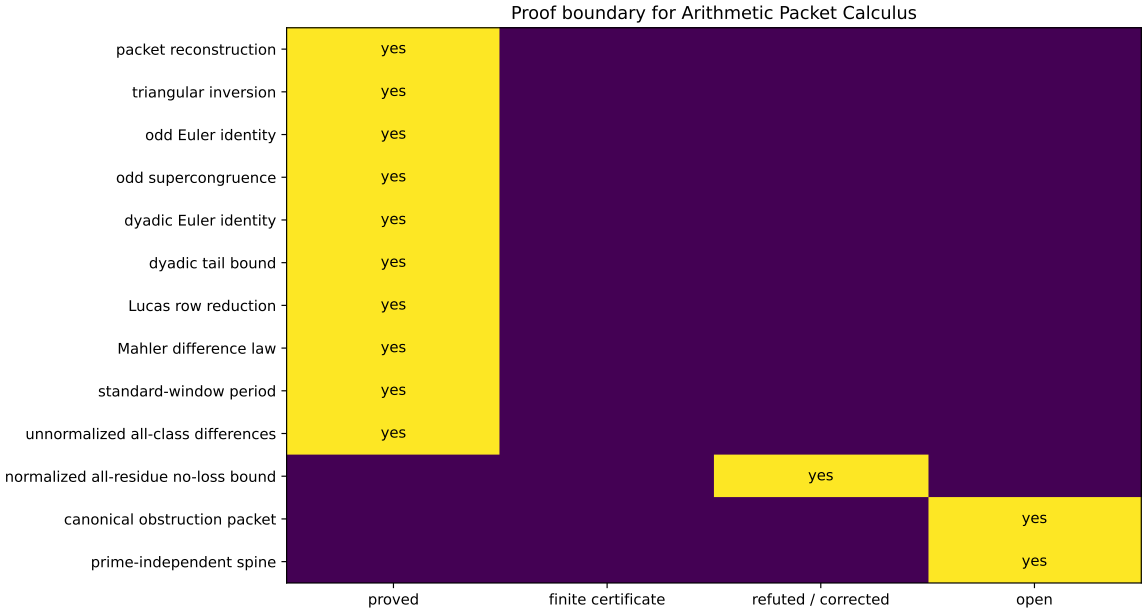


Figure 16: The proof boundary. Finite certificates are not silently promoted to uniform theorems, and historical priority remains external.

12 Formula registry and data boundary

Table 2: Arithmetic formula registry.

ID	Family	Internal status	Boundary
AR-01	Arithmetic residue packets	Proved framework	The selector is classical; the common operator role belongs to the series.
AR-02	Unit EGF coordinate change	Proved	Meaningful only with an explicit structured kernel.
AR-03	Congruential Euler packets	Proved derived family	The normalized and unnormalized conventions differ by $j!$.
AR-04	Odd-power Euler coordinates	Proved	Odd prime and even Euler row.
AR-05	Odd supercongruence ladder	Proved	Modulus follows from the visible p^{2m} filtration.
AR-06	Weighted dyadic Euler coordinates	Proved	Separate construction; not a substitution into AR-04.
AR-07	Dyadic inverse coefficients	Proved 2-integral	Coefficients use an ordinary-power expansion.
AR-08	Mahler difference packets	Finite identities proved	Infinite $\mathbb{Z}_p$ expansion requires standard convergence hypotheses.
AR-09	Standard-window packets	Period theorem proved	The full normalized bound holds for $0 \le j < p$.
AR-10	Unit-mode differences	Proved	Finite $\mathbb{Z}_p$-linear sums.
AR-11	Fibonacci jump packets	Proved	General companion-matrix theorem.
AR-12	Bernoulli boundary	Proved identity	Zeta interpretation is a classical input.
AR-13	Dyadic-odd CRT packets	Proved	Visibility weights can differ by prime.
AR-14	Global congruential Euler differences	Proved with normalization correction	All unnormalized classes satisfy the full bound; normalized classes lose $v_p(j!)$, and the no-loss all-residue statement is false.

The machine-readable registry records definitions, status, and mathematical boundaries. Known canonical sequences such as the Euler zigzag, signed secant, Fibonacci, and Lucas families are used as validation anchors.

13 Open arithmetic problems native to the calculus

Proposition 13.1 (Sharp normalization boundary)**.** *The unrestricted normalized all-residue inequality*

$$v_p\Big(\Delta_h^k C_{mp,j}(n)\Big) \ge \min\{k, mn\}$$

is false when $j!$ is not a p-adic unit. For example, with $(p, m, N, j, n, k) = (3, 2, 6, 3, 1, 1)$,

$$C_{6,3}(1) = -\frac{1}{84}, \qquad C_{6,3}(2) = \frac{703}{5460}, \qquad C_{6,3}(2) - C_{6,3}(1) = \frac{64}{455},$$

so

$$v_3(C_{6,3}(2) - C_{6,3}(1)) = 0 < 1.$$

The corrected factorial-loss bound from the global theorem gives $1 - v_3(3!) = 0$ and is attained exactly in this example.

Proof. The displayed coefficients follow from the exact triangular recurrence. Since neither 64 nor 455 is divisible by 3, the difference has 3-adic valuation zero. The final equality is immediate from $v_3(3!) = 1$. □

Conjecture 13.2 (Canonical arithmetic obstruction packet)**.** Every failed lift from precision p^L to p^{L+1} in a finitely presented arithmetic seed system is carried by a canonically defined finite obstruction packet whose vanishing is necessary and sufficient for continuation.

Conjecture 13.3 (Prime-independent arithmetic spine)**.** For a family admitting compatible 2-adic and odd-adic realizations, there is a smallest common packet spine $\mathfrak{C}$ such that

$$\mathfrak{X} \cong \mathfrak{X}_2^{\mathrm{wt}} \times_{\mathfrak{C}} {\prod_{p>2}}' \mathfrak{X}_p.$$

The common spine should record the prime-independent recurrence, spectral, or Frobenius data, while allowing the local separation weights to differ.

Boundary 13.4 (What Paper IX does not prove). The arithmetic packet calculus does not prove every supercongruence and does not imply unrestricted Galois or automorphic reciprocity. The two remaining native conjectures are the canonical arithmetic obstruction packet and the prime-independent arithmetic spine. Global deformation forests and automorphic occupancy belong to Papers X and XII.

14 Implementation map

A representative exact call is

```
from projector_seed_arithmetic.euler import odd_euler_truncation
from projector_seed_arithmetic.padic import vp_int

exact, truncated, difference = odd_euler_truncation(5, 8, 2)
assert exact == 1385
assert vp_int(difference, 5) >= 6
```

The corresponding dyadic calculation is

```
from projector_seed_arithmetic.euler import dyadic_euler_truncation
from projector_seed_arithmetic.padic import vp_fraction

exact, truncated, difference = dyadic_euler_truncation(8, 2)
assert exact == 1385
assert vp_fraction(difference, 2) >= 6
```

Table 3: Companion implementation.

Module	Responsibility
`packets.py`	Residue projection, exact reconstruction, nested refinement, and CRT addresses.
`euler.py`	Bernoulli, Euler, zigzag, congruential, odd-prime, and weighted dyadic formulas.
`padic.py`	Valuations, Lucas reduction, Mahler coefficients, finite differences, unit decompositions, and finite standard-window certificates.
`recurrences.py`	Companion jumps, Fibonacci and Lucas packets, and packet recurrence residuals.
`experiments.py`	Machine-readable exact certificate bundle.
`visuals.py`	Sixteen code-generated figure pairs.
`tests/`	Thirty-two automated tests across exact packet, transform, valuation, and recurrence identities.

15 Conclusion

The arithmetic realization is not a list of unrelated congruences. It is one operator system:

$$\begin{gathered}\text{coefficient seed} \longrightarrow \text{residue packet} \longrightarrow \text{triangular coordinates} \\ \longrightarrow \text{weighted inverse} \longrightarrow \text{difference packet} \longrightarrow p\text{-adic completion} \\ \longrightarrow \text{exact reconstruction.}\end{gathered}$$

For odd primes, alternating odd-power coordinates give an explicit Euler transform with a two-unit valuation gain at every transfer level. At the dyadic prime, a different coordinate system and inverse are required; their exact generating-function product recovers the same Euler seed, while the natural visible weight is again two per transfer level. Congruential Euler packets place both constructions inside a common triangular family. Lucas reduction explains the first standard window, Mahler coordinates expose finite-difference weights, companion jumps control recurrence packets, and CRT addresses recombine dyadic and odd phases without pretending that their integral denominators are identical.

The result is a proof-normalized arithmetic packet calculus. The global odd-prime finite-difference theorem now separates three distinct facts: the unnormalized all-class bound is exact, the normalized standard-window bound is exact, and the unrestricted normalized all-residue bound fails by a precisely measured factorial loss. The remaining native conjectures concern canonical obstruction packets and prime-independent gluing rather than the finite-difference theorem itself.

A Selected exact rows

A.1 Odd-prime row $n = 8$

For $p = 3$ the depth sequence of omitted-tail valuations is

$$2, 4, 6, 8,$$

which meets the bound at every displayed depth. For $p = 5$ the corresponding values are

$$2, 9, 6, 8.$$

The nonmonotonic middle behavior is harmless: the theorem requires the lower bounds

$$2, 4, 6, 8,$$

not monotonic improvement beyond them.

A.2 Dyadic row $n = 12$

The first five truncated values are

$$\frac{597871}{91}, \quad -\frac{46567427}{273}, \quad \frac{2604301169}{1365}, \quad -\frac{15406834831}{1365}, \quad \frac{47275432817}{1365}.$$

Their error valuations are

$$4, 7, 13, 15, 20,$$

against the uniform bounds

$$2, 4, 6, 8, 10.$$

B Proof-status ledger

Statement	Status	Reason
Arithmetic packet reconstruction	Theorem	Coefficient partition.
Unit EGF coordinate change	Theorem	Exact EGF convolution and inversion.
Congruential triangular inverse	Theorem	Coefficient extraction.
Universal denominator clearing	Proposition	Induction on the triangular row.
Odd-power Euler reconstruction	Theorem	Exact even-part generating function.
Odd-prime supercongruence ladder	Corollary	Every omitted term contains p^{2M+2}.
Dyadic source and inverse integrality	Lemmas	LTE and inversion in $\mathbb{Z}_2[[z^2]]$.
Weighted dyadic reconstruction	Theorem	Exact generating-function product.
Weighted dyadic congruence	Corollary	Every omitted term contains 2^{2M+2}.
Mahler finite-difference formulas	Theorem	Newton interpolation and binomial shifts.
Standard-window Lucas reduction	Theorem	Base-p digit separation.
Standard-window period	Corollary	Follows from the global theorem in the unit-factorial window.
Unnormalized all-class finite-difference bound	Theorem	Complete proof supplied in the formal supplement.
Normalized all-residue bound	Corrected theorem/counterexample	Sharp loss $v_p(j!)$; no-loss version is false beyond the standard window.